

First Normal Form (1NF) Relational Database Design

> **Definition:** A relation is in **First Normal Form (1NF)** if and only if all attributes contain only **atomic (indivisible) values**, and there are no repeating groups or arrays stored within a single field.

1. Detailed Technical Explanation

Normalization is the systematic process of organizing relational database schemas to eliminate data redundancy and anomalies (Insertion, Deletion, and Update anomalies).

Key Rules of 1NF:

1. **Atomic Values:** Each column must store a single value (no multi-valued attributes like comma-separated phone numbers).

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Conversion Example: Unnormalized Table to 1NF

Unnormalized Relation (UNF):

| Stud

Normalized 1NF Relation:

| Stud

2. Core Concepts & Memory Keywords

- **Atomic Domains:** Fields storing single indivisible values.

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3. Must-Write Points for Exams

- 1NF eliminates **multi-valued attributes** and **composite attributes** from relation instances.

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4. Quick Recall Flow

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